

```

In[*]:= (*Second order accurate Schemes for second derivative*)
p[x_] := a + b*x + c*x^2 + d*x^3 (*defined a cubic polynomial*)

ClearAll[x0, xp1, xp2, xp3, eqnsFD, solFD, FDSA];
(*Forward difference*)
x0 = 0;
xp1 = h1;
xp2 = h1 + h2;
xp3 = h1 + h2 + h3;

eqnsFD = { (*Setting up the equations here*)
  p[x0] == fi,
  p[xp1] == fp1,
  p[xp2] == fp2,
  p[xp3] == fp3
};

solFD = Solve[eqnsFD, {a, b, c, d}]; (*solving the equations for the coefficients*)

FDSA = 2 * c /. solFD[[1]]; (*Extracting the required coefficient*)

Simplify[FDSA]

```

```

Out[*]=
(2 (-fp3 h1 h2 (2 h1^2 + 3 h1 h2 + h2^2) + fi h2 h3 (h2 + h3) (3 h1 + 2 h2 + h3) -
  fp1 (h1 + h2) h3 (2 h1^2 + 4 h1 h2 + 2 h2^2 + 3 h1 h3 + 3 h2 h3 + h3^2) +
  fp2 h1 (h2 + h3) (2 h1^2 + 3 h1 (h2 + h3) + (h2 + h3)^2)) /
(h1 h2 (h1 + h2) h3 (h2 + h3) (h1 + h2 + h3))

```

```
In[*]:= ClearAll[x0, xp1, xp2, xm1, eqnsCD, solCD, CDSA];
```

```
(*Central Difference*)
```

```
xm1 = -h1;
```

```
x0 = 0;
```

```
xp1 = h2;
```

```
xp2 = h3 + h2;
```

```
eqnsCD = {
```

```
  p[xm1] == fm1,
```

```
  p[x0] == fi,
```

```
  p[xp1] == fp1,
```

```
  p[xp2] == fp2
```

```
};
```

```
solCD = Solve[eqnsCD, {a, b, c, d}];
```

```
CDSA = 2 * c /. solCD[[1]];
```

```
Simplify[CDSA]
```

```
Out[*]=
```

$$\frac{\left(2 \left(\text{fp2 } h_1 h_2 \left(h_1^2 - h_2^2\right) + \text{fi } (h_1 + h_2) h_3 \left(h_1^2 - h_1 h_2 - 2 h_2^2 - 3 h_2 h_3 - h_3^2\right) + \text{fm1 } h_2 h_3 \left(2 h_2^2 + 3 h_2 h_3 + h_3^2\right) + \text{fp1 } h_1 (h_2 + h_3) \left(-h_1^2 + (h_2 + h_3)^2\right)\right)\right)}{(h_1 h_2 (h_1 + h_2) h_3 (h_2 + h_3) (h_1 + h_2 + h_3))}$$

```
In[*]:= ClearAll[x0, xm1, xm2, xm3, eqnsBD, solBD, BDSA];
```

```
(*Backward Difference*)
```

```
xm1 = -h1;
```

```
x0 = 0;
```

```
xm2 = -h2 - h1;
```

```
xm3 = -h3 - h2 - h1;
```

```
eqnsBD = {
```

```
  p[xm1] == fm1,
```

```
  p[x0] == fi,
```

```
  p[xm2] == fm2,
```

```
  p[xm3] == fm3
```

```
};
```

```
solBD = Solve[eqnsBD, {a, b, c, d}];
```

```
BDSA = 2 * c /. solBD[[1]];
```

```
Simplify[BDSA]
```

```
Out[*]=
```

$$\frac{\left(2 \left(-\text{fm3 } h_1 h_2 \left(2 h_1^2 + 3 h_1 h_2 + h_2^2\right) + \text{fi } h_2 h_3 (h_2 + h_3) (3 h_1 + 2 h_2 + h_3) - \text{fm1 } (h_1 + h_2) h_3 \left(2 h_1^2 + 4 h_1 h_2 + 2 h_2^2 + 3 h_1 h_3 + 3 h_2 h_3 + h_3^2\right) + \text{fm2 } h_1 (h_2 + h_3) \left(2 h_1^2 + 3 h_1 (h_2 + h_3) + (h_2 + h_3)^2\right)\right)\right)}{(h_1 h_2 (h_1 + h_2) h_3 (h_2 + h_3) (h_1 + h_2 + h_3))}$$

```

In[*]:= (*First order accurate second derivative*)
Clear[p];
p[x_] := a + b*x + c*x^2; (*defined a quadractic polynomial*)

ClearAll[x0, xp1, xp2, solFD, eqnsFD, FDFA];
(*Forward difference*)

x0 = 0;
xp1 = h1;
xp2 = h1 + h2;

eqnsFD = {
  p[x0] == fi,
  p[xp1] == fp1,
  p[xp2] == fp2
};

solFD = Solve[eqnsFD, {a, b, c}];

FDFA = 2 * c /. solFD[[1]];

Simplify[FDFA]

```

```

Out[*]=

$$\frac{2 (fp2 h1 + fi h2 - fp1 (h1 + h2))}{h1 h2 (h1 + h2)}$$


```

```

In[*]:= ClearAll[x0, xm1, xp1, solCD, eqnsCD, CDFA];
(*Central difference*)

x0 = 0;
xm1 = -h1;
xp1 = h2;

eqnsCD = {
  p[x0] == fi,
  p[xm1] == fm1,
  p[xp1] == fp1
};

solCD = Solve[eqnsCD, {a, b, c}];

CDFA = 2 * c /. solCD[[1]];

Simplify[CDFA]

```

```

Out[*]=

$$\frac{2 (fp1 h1 + fm1 h2 - fi (h1 + h2))}{h1 h2 (h1 + h2)}$$


```

```
In[*]:= ClearAll[x0, xm1, xm2, solBD, eqnsBD, BDFA];
(*Backward difference*)
```

```
x0 = 0;
xm1 = -h1;
xm2 = -h1 - h2;
```

```
eqnsBD = {
  p[x0] == fi,
  p[xm1] == fm1,
  p[xm2] == fm2
};
```

```
solBD = Solve[eqnsBD, {a, b, c}];
```

```
BDFA = 2 * c /. solBD[[1]];
```

```
Simplify[BDFA]
```

```
Out[*]=
```

$$\frac{2 (fm2 h1 + fi h2 - fm1 (h1 + h2))}{h1 h2 (h1 + h2)}$$

```
In[*]:= (*Second order accurate first derivatives*)
ClearAll[CDSA, FDSA, BDSA];
CDSA = b /. solCD[[1]];
FDSA = b /. solFD[[1]];
BDSA = b /. solBD[[1]];
Simplify[CDSA]
Simplify[FDSA]
Simplify[BDSA]
```

```
Out[*]=
```

$$\frac{fp1 h1^2 - fm1 h2^2 + fi (-h1^2 + h2^2)}{h1 h2 (h1 + h2)}$$

```
Out[*]=
```

$$\frac{-fp2 h1^2 + fp1 (h1 + h2)^2 - fi h2 (2 h1 + h2)}{h1 h2 (h1 + h2)}$$

```
Out[*]=
```

$$\frac{fm2 h1^2 - fm1 (h1 + h2)^2 + fi h2 (2 h1 + h2)}{h1 h2 (h1 + h2)}$$